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Hola!
This notebook is part of a series of geoscience applications designed to demonstrate practical approaches for solving problems using computational methods. It serves as a hands-on guide for understanding key concepts in energy-related subsurface systems through applied problem-solving rather than purely theoretical development. The focus is on building intuition and technical skills for modeling and analysis in real-world contexts, particularly within subsurface energy systems. These include petroleum reservoirs, geothermal resources, carbon storage and sequestration, and mineral exploration systems. By integrating fundamental principles with computational tools, the series aims to support learners in bridging the gap between geoscience theory and engineering practice, enabling more effective analysis and decision-making in complex subsurface environments.

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$$ \textsf{\huge Compressibility factor z of real gases} $$

1. Introduction

In the study of hydrocarbons, it is essential to understand the behavior of gas mixtures, since these fluids are present at various stages of oil reservoir production. Gas analysis is initially based on the theory of ideal gases, which assumes:

1. volume of gas molecules is negligible compared to the volume of the container
2. Intermolecular attractive and repulsive forces are absent
3. Molecular collisions are perfectly elastic, with no net gain or loss of energy

The compressibility factor, z, is defined as the ratio of the actual volume occupied by n moles of a real gas at a given pressure and temperature to the volume that the same amount of gas would occupy if it behaved as an ideal gas under the same conditions:

$$ z=\frac{\text{Volume of real gas @ p and T}}{\text{Volume of ideal gas @ p and T}} $$

The degree to which a gas deviates from ideal behavior depends on pressure, temperature, and composition. Since real gases do not behave ideally, a correction factor known as the gas compressibility factor (z-factor) was introduced to accurately relate pressure, volume, and temperature. By incorporating this factor, the behavior of real gases can be represented more accurately. The resulting equation is given by:

$$ pV = znRT $$

For an ideal gas, the compressibility factor is always equal to 1. In contrast, for a real gas, the compressibility factor can take values either below or above 1, depending on the pressure and temperature conditions. It is important to note that attractive and repulsive forces are the primary mechanisms responsible for deviations in the volumetric behavior of real gases. At low pressures, attractive forces dominate; therefore, the volume of a real gas is lower than that predicted by the ideal gas model. However, at high pressures, repulsive forces become dominant, resulting in compressibility factor values greater than 1.

There are several methods available to compute the compressibility factor, z, some of the most widely used include:

- Standing and Katz
- Sarem
- Brills and Beggs
- Dranchuk and Abou-Kassem (DAK)

2. Numerical solution

2.1 Import libraries

In [9]:
```
import numpy as np
import matplotlib.pyplot as plt
```

2.2 Define DAK coefficients

In [ ]:
```
# DAK coefficients
A1  = 0.3265
A2  = -1.0700
A3  = -0.5339
A4  = 0.01569
A5  = -0.05165
A6  = 0.5475
A7  = -0.7361
A8  = 0.1844
A9  = 0.1056
A10 = 0.6134
A11 = 0.7210
```

2.3 Function for F calculation

In [2]:
```
def dak_F(z, Pr, Tr):
    rho_r = 0.27 * Pr / (z * Tr)

    term1 = (A1 + A2/Tr + A3/Tr**3 + A4/Tr**4 + A5/Tr**5) * rho_r
    term2 = (A6 + A7/Tr + A8/Tr**2) * rho_r**2
    term3 = A9 * (A7/Tr + A8/Tr**2) * rho_r**5
    term4 = A10 * (1 + A11 * rho_r**2) * (rho_r**2 / Tr**3) * np.exp(-A11 * rho_r**2)

    return z - (1 + term1 + term2 - term3 + term4)
```

2.4 Function for derivative of F

In [3]:
```
def dak_dFdz(z, Pr, Tr):
    rho_r = 0.27 * Pr / (z * Tr)

    C1 = (A1 + A2/Tr + A3/Tr**3 + A4/Tr**4 + A5/Tr**5)
    C2 = (A6 + A7/Tr + A8/Tr**2)
    C3 = (A7/Tr + A8/Tr**2)

    term1 = C1 * (rho_r / z)
    term2 = 2 * C2 * (rho_r**2 / z)
    term3 = -5 * A9 * C3 * (rho_r**5 / z)

    exp_term = np.exp(-A11 * rho_r**2)
    bracket = (1 + A11 * rho_r**2 - (A11 * rho_r**2)**2)
```

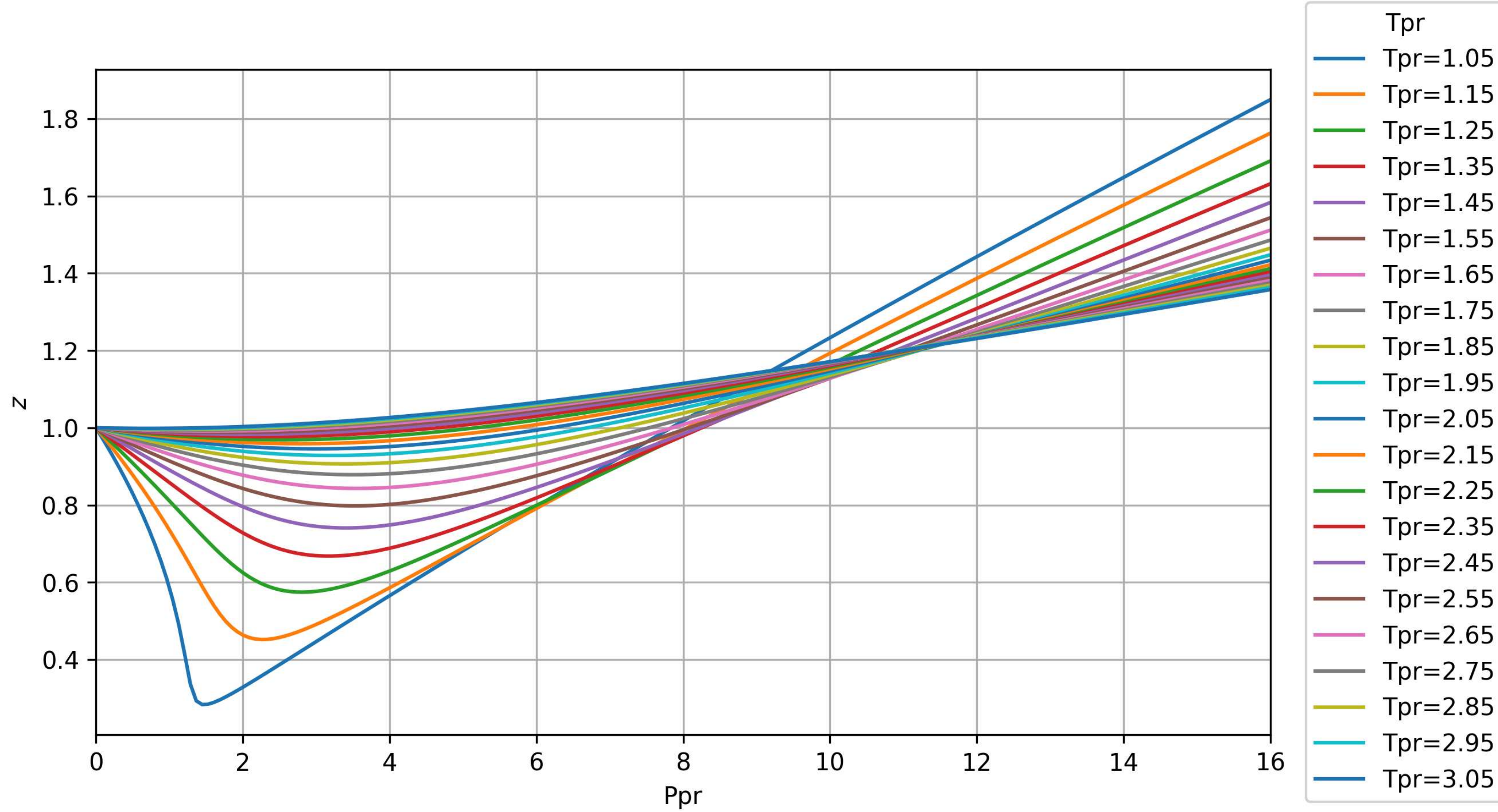
```python
    term4 = (2 * A10 * rho_r**2 / (z * Tr**3)) * bracket * exp_term

    return 1 + term1 + term2 + term3 + term4
```

Function to solve for z using Newton-Raphson

In [4]:
```python
def solve_z_newton(Pr, Tr, z0, tol=1e-4, max_iter=100):
    z = z0

    for i in range(max_iter):
        F = dak_F(z, Pr, Tr)
        dF = dak_dFdz(z, Pr, Tr)

        z_new = z - F / dF

        if abs(z_new - z) < tol:
            return z_new, i+1

        z = z_new

    raise RuntimeError("Newton-Raphson did not converge")
```

2.6 Plotting

In [5]:
```python
# Example
# Pr = 5.60
Pr = np.linspace(0, 16, 200)
Tr = np.linspace(1.05,3.05,21)

plt.figure(dpi=300, figsize=(9,5))

for j in range(len(Tr)):
    z_vals = []  # store z for this Tr

    for i in range(len(Pr)):
        z, iterations = solve_z_newton(Pr[i], Tr[j], 1)
        z_vals.append(z)

    # plot full line for this Tr
    # plt.plot(Pr, z_vals, label=f'Tr={Tr[j]:.2f}')
    plt.plot(Pr, z_vals,label=f'Tpr={Tr[j]:.2f}' )


plt.xlim(0, 16)
plt.xlabel('Ppr')
plt.ylabel('$z$')
plt.grid()

# --- Legend outside with box ---
plt.legend(
    loc='center left',
    bbox_to_anchor=(1.02, 0.5),  # move outside right
    frameon=True,                # draw box
    title='Tpr'
)

# Adjust layout so nothing is cut off
plt.tight_layout()

plt.show()
```



3. Conclusions and highlights

In [ ]:

4. Comments

I hope this notebook is useful in your journey of coding appplied to petroleum engineering and geosciences. I am a researcher interested in Machine learning for subsurface energy systems including oil & gas, carbon capture and storage, geothermal, and minerals. Please reach out!

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```
In [ ]:
```